

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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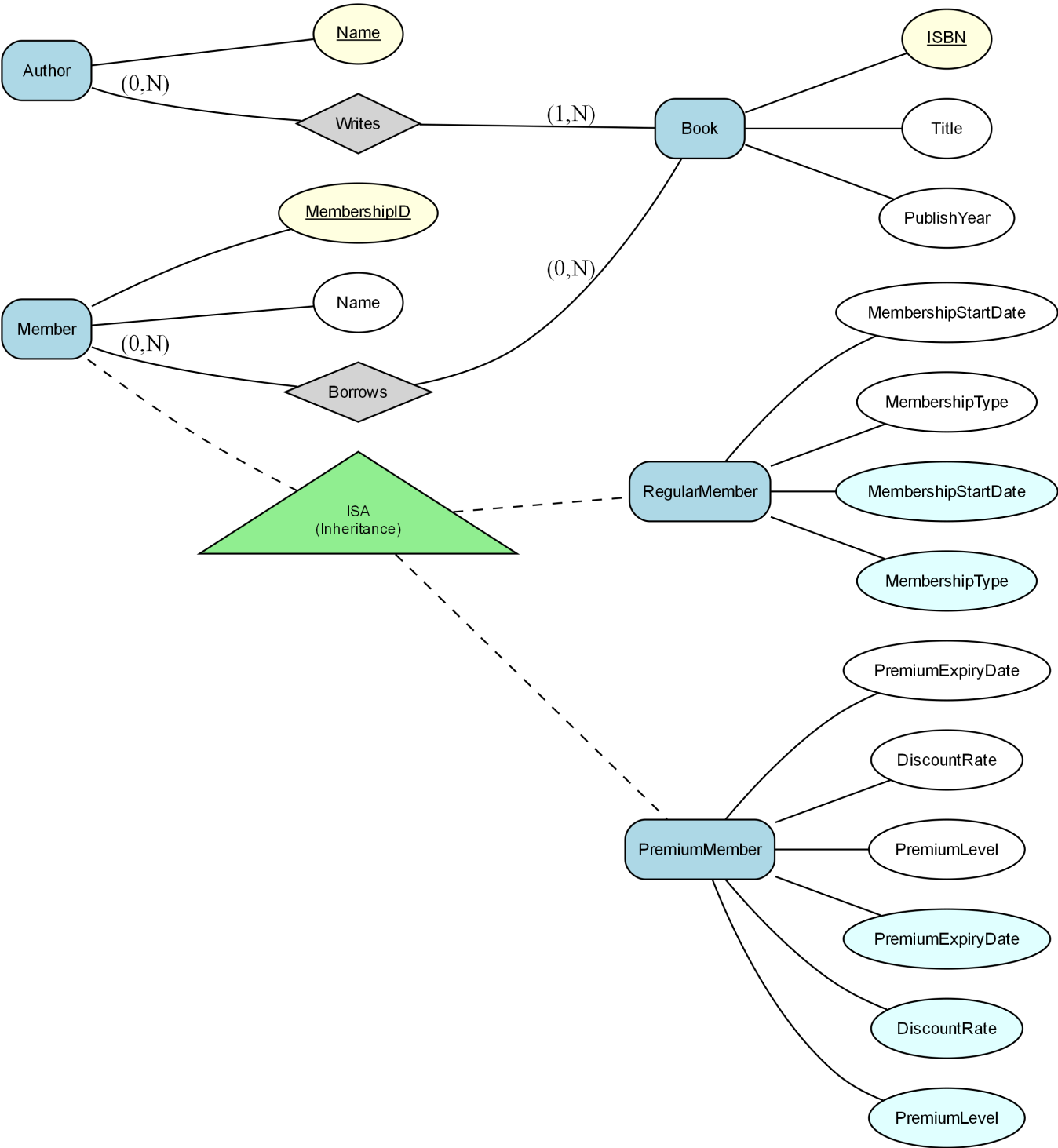
Total Marks: 100

Examination Paper

Question Q1 (20 marks)

Consider the following Enhanced Entity-Relationship (EER) diagram:

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys

and the foreign keys of the resulting relations clearly. (17)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3)

Question Q2 (15 marks)

Consider a relation $R(\text{EmployeeID}, \text{EmployeeName}, \text{DepartmentID}, \text{DepartmentName})$ with the following set of functional dependencies F over R : $F = \{\text{EmployeeID} \rightarrow \text{EmployeeName}, \text{DepartmentID} \rightarrow \text{DepartmentName}, \text{EmployeeID} \rightarrow \text{DepartmentID}\}$.

a) Draw the functional dependency diagram. (1)

b) Use the attribute closure and identify the primary key for the given relation. (1)

c) If we insert a row $\langle \text{E123}, \text{John Doe}, \text{D1}, \text{null} \rangle$ into the given relation, what is the update anomaly that is violated? Give reasons. (2)

d) Assume that the relation has a row, $\langle \text{E124}, \text{Jane Smith}, \text{D2}, \text{Logistics} \rangle$. If we remove the department D2, what is the possible update anomaly that can be violated? Give reasons. (2)

e) Using the above functional dependencies, design a set of 3NF relations for the above given relation. Show clearly each stage in deriving the 3NF relations. Also identify the normal form at each of these stages. (3)

f) Do the update anomalies of part (c) and (d) hold in the 3NF relations? Discuss briefly. (2)

g) Are the 3NF relations derived in part (e) still in 3NF? Justify your answer. (2)

h) Are the 3NF relations derived in part (e) in BCNF? If not, bring the relations to BCNF. (2)

Question Q3 (25 marks)

A company named TechSolutions is setting up a new database system to manage its inventory of products and suppliers. The database schema will include tables for Products (product_id : integer, product_name : string, price : real) and Suppliers (supplier_id : integer, supplier_name : string, city : string). The database team is responsible for implementing various operations on this schema. As part of their tasks, they need to ensure data integrity and proper access controls.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2)

b) Code Segment:

```
```sql
PreparedStatement pstmt = connection.prepareStatement(
 "SELECT * FROM Employees WHERE Department = ?");
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();
```
```

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2)

c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2)

d) Briefly explain how authentication and authorization are achieved in SQL Server. (3)

e) A firm is developing a database system to manage its product inventory. Sarah is the lead database administrator tasked with maintaining the entire database system. Tom is a junior database administrator responsible for the Products table while Emily manages data input for the Suppliers table. Write a T-SQL statement to create a login to Sarah with windows authentication. (3)

f) Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3)

g) Assuming Emily's username is 'emily.s', write T-SQL statements to assign her the responsibility of managing the Suppliers table using a user-defined server role. (4)

h) Assuming Tom's login name is 'tom.t', write T-SQL statements to grant him permission to create tables within the inventory database. (3)

i) Assuming the company assigns data entry responsibilities to Michael with login name 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Products and Suppliers tables. (3)

Question Q4 (40 marks)

Consider the following relational database schema containing details about a library system. In this system, the book relation stores information about different books, the member relation holds details about library members, the transaction relation records borrowings and returns, and the author relation contains information about book authors.

book (bid, title, genre, author_id)

member (mid, name, email, phone)

transaction (tid, mid, bid, date, status)

author (author_id, author_name, country)

- a) Find the titles of books that have been borrowed at least once. (5)
- b) Identify the members who have never borrowed any books from the library. (5)
- c) Get the details of authors who have written books in the 'Science Fiction' genre. (5)
- d) Retrieve the names of members who have borrowed books by a specific author_id. (5)
- e) How many books are currently available (not borrowed)? (5)
- f) Express the above queries (a) to (d) in tuple calculus. (15)